

MAYANK ARAMBHI

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SUMMARY

Computer Engineering and Data Science student, building desktop apps, interactive games, and data visualization projects. Interested in backend development, machine learning and scalable software engineering.

EDUCATION

- Indian Institute of Technology Madras**
BS in Data Science and Applications, **9 CGPA** (May 2025 – Present)
- Government Engineering College, Gandhinagar**
BE in Computer Engineering (Aug 2025 – Present)

EXPERIENCE

Spaceborn – Data Engineering Intern (Feb 2025 – Present)

FastAPI, Python, Plotly, Pandas

- Built a real-time telemetry dashboard handling 10Hz drone telemetry streams with WebSocket broadcasting to multiple clients.
- Developed FastAPI backend services for live data streaming.
- Optimized backend communication pipelines for scalable telemetry streaming.

PROJECTS

Weather Wall | [GitHub](#)

C#, .NET Framework, WPF, Weather APIs

- Built a lightweight Windows desktop utility that dynamically changes wallpapers using live weather, local time, and atmospheric conditions.
- Developed an AI-assisted wallpaper matching pipeline that maps wallpapers to the closest weather/time conditions from user-selected libraries.
- Implemented automated wallpaper synchronization, manual override workflows, and optimized background operation with ~5–12 MB idle RAM usage.

Earth Elevation Analysis & Visualization | [GitHub](#)

Python, NumPy, OpenCV, Matplotlib, Scikit-learn, SciPy

- Processed and analyzed a global relief map containing **10M+ pixels** using Python-based image processing and terrain extraction techniques.
- Applied **K-Means clustering (55 clusters)** and nonlinear elevation mapping to reconstruct approximate terrain distribution patterns.
- Generated smoothed global elevation distribution curves using **Matplotlib, NumPy, and SciPy**, including latitude-based weighting corrections.
- Built a complete data-processing pipeline for extracting, classifying, clustering, and visualizing terrain characteristics from rasterized elevation maps.

Survive the Balls | [GitHub](#)

Python, Pygame

- Built a real-time 2D arcade survival game using the **Pygame** framework.
- Implemented collision-detection systems, dynamic enemy spawning, and progressive difficulty scaling mechanics.
- Developed physics-based bouncing behavior, audio effects, survival-time tracking, and persistent high-score logging.

SKILLS

Languages: Python, C

Backend: FastAPI, WebSockets

Data & Visualization: Pandas, NumPy, Matplotlib

Frameworks & Tools: WPF, Git, GitHub, VS Code

Concepts: Real-Time Streaming, Desktop Applications, Data Processing

ACHIEVEMENTS & CERTIFICATIONS

- Participated in Lakshya 2.0 Hackathon at LD College of Engineering (LDCE)
- Completed Pandas and Data Visualization courses on Kaggle